TIM-Rec: An Industrial Dataset for Multi-Item Upselling with Sparse Feedback*

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Abstract

Upselling recommendations in telecommunications are critical for customer engagement and revenue, yet real-world interactions feature multi-item suggestions per session and highly sparse explicit feedback, complicating model evaluation. To bridge the critical gap in public resources capturing these industrial complexities, we introduce **TIM-Rec**, a novel dataset capturing over 1.5M anonymized interactions. TIM-Rec records multiple campaign-specific offers presented during telephone calls alongside explicit user responses. We provide comprehensive dataset statistics and establish performance benchmarks across diverse recommendation models, ranging from classical matrix factorization to transformer-based sequential architectures. Our findings highlight the unique challenges of high-sparsity industrial environments.

Keywords

Recommender systems, Dataset, Telecommunication

TIM-Rec Dataset Card

Repositories: Hugging Face Dataset | GitHub Code
Industry: Telecommunications
Statistics: 1.5M Interactions | 370k Users | 388 Items | 6 Months
Recommendation Setting: Multiple offers presented simultaneously per call
Maintenance Status: Static dataset (monitored for technical issues)

1. Introduction

Upselling encourages customers to purchase higher-value upgrades or enhancements, maximizing transaction value while addressing evolving customer needs [2, 3]. In telecommunications, upselling is a key driver for increasing the Average Revenue Per User (ARPU) by transitioning clients to premium plans or incorporating value-added services (e.g., streaming or security options) [4, 5]. While personalization through recommender systems is highly effective [6], developing robust models is severely hindered by the scarcity of publicly available, telco-specific datasets, as existing public resources focus almost exclusively on e-commerce or media domains [7, 8].

To bridge this gap, we introduce **TIM-Rec**, a novel, public, real-world dataset comprising over 1.5 million interactions from actual customer-agent phone calls in a major telecommunications company. TIM-Rec features two key industrial complexities rarely captured in existing literature: (i) *Multi-item recommendations* (Agents suggest multiple offers simultaneously within a single session) and (ii) *Sparse explicit feedback* (Customers frequently decline offers or end calls without feedback, leading to extremely low acceptance rates (5%)).

The main contributions of this work are twofold:

- **Dataset Release**: We provide a large-scale, anonymized, and privacy-compliant dataset containing 1.5M interactions, 370k users, and 388 items, complete with fine-grained contextual features

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reduced via PCA.

- **Evaluation Benchmark**: We adapt well-known LTR metrics to handle sparse explicit feedback and benchmark a wide range of paradigms (from classical matrix factorization to sequential transformer architectures) to establish strong baselines.

2. Related Work

Dataset	Interactions	Time Range	Dataset Type	Acceptance Sparsity (%)	# Rec.	Public	Dynamic
Amazon [9]	∼571M	27 years	Cross-selling, Upselling	Not Applicable	∼ 1.0	✓	×
Movielens25M [7]	25M	24 years	Recommendation	Not Applicable	∼ 1.2	✓	×
Taobao [10]	∼100M	9 days	Upselling	∼ 2% (implicit)	∼ 1.0	✓	✓
MIND [11]	∼17.7M	6 weeks	Recommendation	∼11% (implicit)	∼ 37	✓	✓
Iranian Churn [12]	∼3k	1 year	Customer Churn	Not Applicable	1	✓	×
Dookeram et al. [13]	∼3k	1 year	Upselling	Not Applicable	?	×	✓
Alves et al. [14]	∼8M	?	Recommendation	Not Applicable	?	×	×
Pham et al. [15]	∼80k	6 months	Recommendation	Not Applicable	?	×	✓
Ours	∼1.5M	6 months	Upselling	**∼5% (explicit)**	**∼2.4**	✓	✓

Table 1
Comparison of TIM-Rec against general and telecom-specific datasets. The "Dynamic" column indicates whether interactions were recorded in real time (✓) or collected retrospectively (×).

Commonly used datasets for collaborative filtering and Learning-to-Rank (LTR), such as MovieLens [7], Amazon [9], and Netflix [8], focus primarily on e-commerce or media entertainment [11, 10]. These benchmarks typically assume single-item interactions with implicit signals [10] or static rating scales [7], which fail to capture the dynamic, multi-item recommendation setting and high-sparsity explicit feedback characteristics of industrial telecommunications upselling [16, 17].

Public datasets in the telecommunications domain are notoriously scarce [18]. The Iranian Churn dataset [12] spans 12 months but provides only aggregate, churn-focused statistics rather than granular upselling interactions. Other studies targeting telecom upselling either rely on very small datasets (e.g., 3k samples) [13] or utilize large-scale datasets (from 8M to 80M interactions) that remain strictly proprietary and closed-source [14, 15].

As shown in Table 1, **TIM-Rec** is the first large-scale, public, and dynamic industrial dataset specifically designed to study multi-item upselling under explicit, highly-sparse negative and positive feedback.

3. Dataset Overview

3.1. Dataset Construction

Collected over a 180-day period from real-world inbound telecommunications interactions, **TIM-Rec** contains 370,314 users, 388 items, and 1,494,061 interactions [1]. To isolate upselling dynamics, we applied three key filtering steps:

Campaign Filtering: Churn-prevention (retention) and cross-selling instances were removed to focus strictly on campaign-specific upselling proactive offers.

Call Abandonment: Abandoned calls (over 99% of raw interactions) were excluded. Their feedback is highly ambiguous (e.g., dropped lines or hang-ups before an offer is made); treating them as negative responses would severely skew and over-penalize recommendation models.

Representative Undersampling: We applied random user-level undersampling via PySpark to scale down the dataset while preserving behavioral and attribute distributions.

3.2. Preliminaries and Privacy Safeguards

Each interaction is represented as a tuple $(x_i, y_i, d_i, c_i, f_i, \vec{v}_i)$, where x_i is the anonymized user ID, y_i the item ID, d_i the contact date, c_i is the item category, $f_i \in \{\text{Accepted, Refused}\}$ the explicit feedback, and $\vec{v}_i \in \mathbb{R}^{64}$ the user's contextual feature vector. To guarantee strict privacy, we collect data only from users who provided explicit consent, desensitize IDs, and compress 550 raw behavioral and demographic features (including billing, network usage, ARPU, and customer care history) into a 64-dimensional vector using Principal Component Analysis (PCA) [1]. TIM-Rec is published under the CC-BY-NC 4.0 license.

3.3. Dataset Statistics and Sparsity Patterns

Table 2 summarizes the dataset statistics. A defining aspect of TIM-Rec is its *multi-item recommendation setting*: each call contains a minimum of 2 recommended items (averaging 2.428). Furthermore, positive feedback is extremely sparse: only 5.286% of calls result in at least one accepted offer, and a mere 0.233% achieve three or more acceptances, reflecting the high-sparsity nature of real-world upselling. Temporally, acceptance rates show significant seasonal variations, peaking in June and July (up to 20.90%), likely due to targeted summer marketing campaigns. User intercontact times exhibit a strong leftward skew, indicating that user engagement occurs in rapid bursts, while item re-engagements are more widely distributed over periodic cycles.

Table 2: Summary of TIM-Rec Dataset Statistics.

Metric	Value
Unique Calls	613,449
Unique Users	370,314
Unique Recommendations	1,494,061
Unique Items	388
Avg. Recs per Call	2.428
Avg. Acceptances per Call	0.153
Calls with $\geq$ 1 Acceptance	5.286%
Calls with $\geq$ 2 Acceptances	1.373%
Calls with $\geq$ 3 Acceptances	0.233%

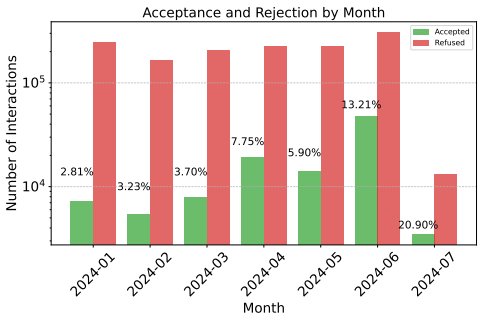


Figure 1: Acceptance sparsity per month.

4. Experimental Setup

4.1. Dataset Splitting

To prevent temporal cold-start issues caused by the monthly introduction of new offers, we partition the dataset by user and time. For each user, we select their last 20% of interactions as the test set, the preceding 10% for validation, and the remaining 70% for training. This random user-based split preserves behavior diversity while ensuring a balanced and representative distribution across splits.

4.2. Evaluation Metrics and Modified NDCG

We evaluate models using Precision@K, Recall@K, and NDCG@K ($K \in \{5, 10, 20\}$). Because explicit negative feedback is abundant and positive feedback is sparse, standard metrics do not fully capture recommendation effectiveness. Thus, we modify the relevance score in the standard NDCG@k formula to handle unseen items alongside explicit responses, assigning 0.5 to unseen offers. We chose 0.5 as we assume an unseen item holds some potential relevance: more than an explicit refusal but less than an acceptance. This models preference uncertainty, encouraging the system to propose novel items rather than items the user has already explicitly rejected.

4.3. Baselines and Training Strategies

We benchmark seven baseline models spanning multiple paradigms: classical matrix factorization (**SVD** [19] and **ALS-WR** [20]), neural collaborative filtering (**NCF** [21]), convolutional sequence recommendation (**Caser** [22]), recurrent networks (**GRU4Rec** [23]), self-attention transformers (**SASRec** [24]), and graph neural networks (**LightGCN** [25]).

To ensure fairness, we fix the latent embedding dimension to 64 across all models. Matrix factorization methods train on all historical labels (SVD until convergence; ALS-WR for 50 iterations with regularization of 0.001). Neural architectures receive the most recent 45 interactions per user, optimizing binary cross-entropy loss via the Adam optimizer (learning rate of 0.001) in PyTorch [26].

5. Results

The benchmark results of the evaluated models are reported in Table 3.

Table 3
Model performance evaluation. **Bold** indicates the best results, while second best is underlined.

Method	N@5	P@5	R@5	N@10	P@10	R@10	N@20	P@20	R@20
SVD	0.073	0.022	0.083	0.074	0.012	0.089	0.075	0.006	0.091
ALS-WR	0.150	0.049	0.219	**0.205**	**0.043**	**0.388**	**0.220**	**0.024**	**0.444**
NCF	**0.164**	**0.053**	**0.246**	0.173	0.029	**0.388**	0.192	0.019	0.347
Caser	0.091	0.032	0.145	0.122	0.026	0.238	0.135	0.016	0.287
GRU4Rec	0.036	0.016	0.071	0.062	0.017	0.155	0.107	0.018	0.326
SASRec	0.154	0.047	0.218	0.169	0.029	0.264	0.184	0.018	0.325
LightGCN	0.080	0.028	0.131	0.102	0.021	0.198	0.118	0.014	0.260

Analysis of Findings. ALS-WR consistently outperforms nearly all baselines across the majority of metrics (e.g., reaching NDCG@10 0.205, Recall@10 0.388), closely followed by **NCF** and **SASRec**. This superior performance highlights that matrix factorization with weighted regularization is exceptionally robust under severe feedback imbalance and explicit negative signals. NCF proves to be a strong nonlinear competitor, achieving the best overall score on the shortest list length (NDCG@5 0.164).

Among sequential models, **SASRec** performs on par with ALS-WR, confirming that self-attention mechanisms effectively capture long-term customer behavioral dependencies over telephone interactions. Conversely, other deep learning models (**GRU4Rec**, **Caser**, and **LightGCN**) underperform significantly, with GRU4Rec obtaining a low NDCG@5 of 0.036. This performance gap stems from their standard formulations, which are intrinsically optimized for implicit feedback and struggle to incorporate the rich, explicit negative responses characteristic of the TIM-Rec dataset.

6. Conclusion and Future Work

In this work, we introduced **TIM-Rec**, a novel, large-scale industrial dataset designed to study multi-item upselling under severe, explicit feedback sparsity in the telecommunications sector. By releasing over 1.5 million anonymized interactions and establishing a robust benchmarking pipeline, we bridge a critical gap in publicly available recommender systems resources. Our experimental results reveal that traditional matrix factorization with weighted regularization significantly outperforms other deep learning models. This performance gap underscores the limitations of standard neural architectures, which are natively optimized for implicit signals and struggle to accommodate explicit negative feedback. Future research will focus on developing specialized loss functions to handle explicit rejections and investigating reinforcement learning frameworks for dynamic, real-time upselling strategies during live customer-agent calls.

Declaration on Generative AI

During the preparation of this work, the authors used generative AI tools in order to improve writing style and Grammar and spelling check. After using these tools, the authors reviewed and edited the content as needed and takes full responsibility for the publication's content.

References

[1] A. Sbandi, F. Siciliano, F. Silvestri, Tim-rec: Explicit sparse feedback on multi-item upselling recommendations in an industrial dataset of telco calls, in: Proceedings of the Nineteenth ACM Conference on Recommender Systems, 2025, pp. 865–873.

[2] M. Pandya, A. Dholakia, Upselling strategy: A review, Towards Excellence 13 (2021).

[3] D. Rauhaus, J. Gönsch, C. Steinhardt, On the value of booking data for upsell decision-making in revenue management, Flexible Services and Manufacturing Journal (2024) 1–34.

[4] A. Lambrecht, B. Skiera, Paying too much and being happy about it: Existence, causes, and consequences of tariff-choice biases, Journal of marketing Research 43 (2006) 212–223.

[5] K. Pauwels, A. Weiss, Moving from free to fee: How online firms market to change their business model successfully, Journal of Marketing 72 (2008) 14–31.

[6] J. Dietz, Complaint management: The heart of crm, Journal of Consumer Marketing 23 (2006) 50–51.

[7] F. M. Harper, J. A. Konstan, The movielens datasets: History and context, Acm transactions on interactive intelligent systems (tiis) 5 (2015) 1–19.

[8] J. Bennett, S. Lanning, et al., The netflix prize, in: Proceedings of KDD cup and workshop, volume 2007, New York, 2007, p. 35.

[9] J. Ni, J. Li, J. McAuley, Justifying recommendations using distantly-labeled reviews and fine-grained aspects, in: Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP), 2019, pp. 188–197.

[10] G. Zhou, N. Mou, Y. Fan, Q. Pi, W. Bian, C. Zhou, X. Zhu, K. Gai, Deep interest evolution network for click-through rate prediction, in: Proceedings of the AAAI conference on artificial intelligence, volume 33, 2019, pp. 5941–5948.

[11] F. Wu, Y. Qiao, J.-H. Chen, C. Wu, T. Qi, J. Lian, D. Liu, X. Xie, J. Gao, W. Wu, et al., Mind: A large-scale dataset for news recommendation, in: Proceedings of the 58th annual meeting of the association for computational linguistics, 2020, pp. 3597–3606.

[12] Iranian Churn, UCI Machine Learning Repository, 2020. DOI: https://doi.org/10.24432/C5JW3Z.

[13] N. Dookeram, Z. Hosein, P. Hosein, A recommender system for the upselling of telecommunications products, in: 2022 24th International Conference on Advanced Communication Technology (ICACT), IEEE, 2022, pp. 66–72.

[14] D. Alves, B. Valente, R. Filipe, M. Castro, L. Macedo, A recommender system for telecommunication operators' campaigns, in: Progress in Artificial Intelligence: 19th EPIA Conference on Artificial Intelligence, EPIA 2019, Vila Real, Portugal, September 3–6, 2019, Proceedings, Part II 19, Springer, 2019, pp. 83–95.

[15] C. D. Pham, T. A. Chu, H. H. Pham, M. L. Dao, T. S. Pham, D. H. Nguyen, et al., A recommendation system for offers in telecommunications, in: 2020 IEEE Eighth International Conference on Communications and Electronics (ICCE), IEEE, 2021, pp. 302–306.

[16] Y.-c. Fan, Y. Ji, J. Zhang, A. Sun, Our model achieves excellent performance on movielens: What does it mean?, ACM Transactions on Information Systems 42 (2024) 1–25.

[17] F. B. Pérez Maurera, M. Ferrari Dacrema, P. Castells, P. Cremonesi, Impression-aware recommender systems, ACM Transactions on Recommender Systems 3 (2025) 1–46.

[18] Z. Lian-Ying, D. M. Amoh, L. K. Boateng, A. A. Okine, Combined appetency and upselling prediction scheme in telecommunication sector using support vector machines, International Journal of Modern Education and Computer Science 14 (2019) 1.

[19] Y. Koren, R. Bell, C. Volinsky, Matrix factorization techniques for recommender systems, Computer 42 (2009) 30–37.
[20] Y. Hu, Y. Koren, C. Volinsky, Collaborative filtering for implicit feedback datasets, in: 2008 Eighth IEEE international conference on data mining, Ieee, 2008, pp. 263–272.
[21] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, T.-S. Chua, Neural collaborative filtering, in: Proceedings of the 26th international conference on world wide web, 2017, pp. 173–182.
[22] J. Tang, K. Wang, Personalized top-n sequential recommendation via convolutional sequence embedding, in: Proceedings of the eleventh ACM international conference on web search and data mining, 2018, pp. 565–573.
[23] B. Hidasi, Session-based recommendations with recurrent neural networks, arXiv preprint arXiv:1511.06939 (2015).
[24] W.-C. Kang, J. McAuley, Self-attentive sequential recommendation, in: 2018 IEEE international conference on data mining (ICDM), IEEE, 2018, pp. 197–206.
[25] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, M. Wang, Lightgcn: Simplifying and powering graph convolution network for recommendation, in: Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, 2020, pp. 639–648.
[26] F. Betello, A. Purificato, F. Siciliano, G. Trappolini, A. Bacciu, N. Tonellotto, F. Silvestri, A reproducible analysis of sequential recommender systems, IEEE Access (2024).